

Sender–Receiver Exercise X: Instructions

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One of the skills we want you to develop in CS1200 is to be able to manage the complexity of mathematical proofs by thinking about them at different levels of abstraction, ranging from a high-level intuitive proof outline to being able to work out the low-level formal details. Importantly, these are not two separate skills—rather you need to learn to be able to *move between* these different levels of thinking about proofs, e.g., extracting the high-level intuition from a formal proof, and turning a high-level proof strategy into a detailed formal proof. The reason this skill is particularly valuable for you as computer scientists is that software and hardware systems can be extremely complex to design and understand; to manage that complexity, we need to be able to work at a high level, but to complete an implementation or analysis, we also need to be able to fill in the technical details.

To develop this skill, we will do half-hour in-class active learning exercises where half of you (the “Senders”) are tasked with reading and understanding a formal proof, and then explaining it a high level to a classmate (a “Receiver”), who is then tasked with writing it down and filling in the formal details. In addition, these exercises will also have the benefit of reinforcing important concepts related to the content of the course.

The structure of the exercise is as follows. *Before class*, both the Sender and Receiver should study the reading assigned to them. The Receiver reading is short and only contains the theorem statement and motivation. The Sender reading contains a proof of the theorem and should be studied more carefully. If you are a Sender, you should try to extract both the main ideas (low-to-high level translation) and the technical details. You may create some bullet notes or a high-level summary for yourself to help you in your presentation during class. Avoid the temptation to write a fully detailed script or to memorize the proof; instead try to internalize the ideas so that you can reproduce the proof based on an intuitive understanding.

The exercise will proceed as follows:

1. We will mark alternating seats as designated for Senders or for Receivers using colored papers or paper flags. (The classroom is full enough that we don’t have room to leave an empty seat between sender-receiver pairs.) When you arrive, sit in a seat designated for Sender or for Receiver, and pair up with the person with opposite designation next to you—ordinarily, receiver on the right, sender on the left. Fill up seats starting at the front of the room, and try to pair up with someone new (that you haven’t paired with in a previous SRE).
2. Hold your colored sheet or flag in the air if you don’t have a partner for the exercise, and don’t put it down until you are paired up. Since we usually will not end up with exactly the same number of Senders and Receivers, some groups may end up as triples.
3. Once the exercise starts, the Sender describes the theorem and proof to the Receiver through an oral, interactive dialogue. The Sender should avoid writing, but may draw a diagram or two to get the explanation started (but further manipulations of the diagram should be left to the Receiver). The Receiver should be capturing their understanding by writing pictures and notes (which can be on the Receiver instructions document, which we will distribute both

as hardcopies and as a pdf the course schedule). The Receiver should ask questions of the Sender during the interaction. After the high-level understanding is transmitted, the dialogue should continue on to filling in as many formal details as possible.

4. Senders and Receivers should raise their hands if they are jointly stuck or have clarification questions, and a member of the teaching staff will come to help.
5. After the exercise is complete (or at the end of class), we will have you fill out a reflection survey, which will provide valuable feedback for us and (for receivers) is part of your participation grade.

We encourage you to be creative and experiment with different strategies for how to make this exercise as effective as possible. Some things that students have found useful in the past are:

- Good pre-class preparation by both the Sender and Receiver.
- Arriving to class on time so that you have full time for the exercise.
- Making sure that the in-class interaction is a two-way dialogue, with questions and suggestions from the Receiver, rather than a one-way explanation.
- A top-down approach to explanation, so that the Receiver knows where the argument is going and how the pieces are relevant to the end goal.
- A judicious use of diagrams drawn by the Sender or the Receiver.
- Trying to think of solutions/proofs on one's own before going into the presented one.